2 Fig. 2.1 shows a slope at 30° to the horizontal. P is a point on a slope and Q is the point the slope meets the horizontal ground. PQ is 1.2 m.

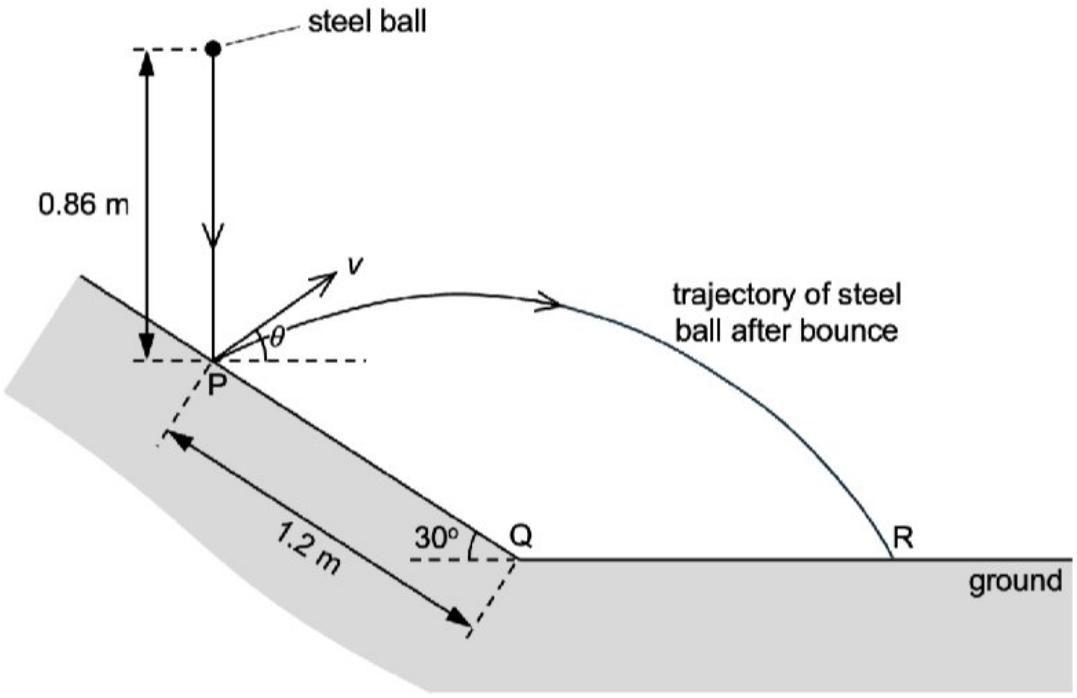


Fig. 2.1

A small steel ball is dropped from a height of 0.86 m above P.

The steel ball collides with the slope at P and bounces off elastically.

(a) Show that the speed of the steel ball just before the collision at P is 4.1 m s^{-1}.

[1]

(b) State the magnitude of the ball's velocity v immediately after the collision.

..... [1]

(c) The velocity v is at an angle θ to the horizontal as shown in Fig. 2.1.

Explain why $\theta = 30°$.

.....

.....

.....

.....

..... [3]

(d) Determine the distance QR.

QR = ... m [3]

[Total: 8]